

Open mobility data for functional-area maps

What each source leaves out, and how much of the result is scale, contiguity and algorithm.

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Prepared for FOSS4G 2026, Hiroshima. Talk title: Eurostat vs OSM vs Census: Choosing Open Mobility Data for Urban Function Maps. The talk was accepted; I was unable to present it. These are the slides in full.

This deck is version 2. Every number in it is written by the pipeline at github.com/m-erts/urban-functional-maps.

■ Eurostat-type grid and presence statistics ■ OpenStreetMap ■ Census and register counts ■ Method, reference

Each open source publishes some of three things

Pairs, purpose, time. The missing one is not an error in the file. It is found in the code list and the definitions.

Source	Origin-destination pairs	Trip purpose	Time of day or week	What the name suggests
■ Census, England and Wales 2011, 2021	full matrix, 7,264 units	work	none	a commuting matrix. In 2021, 45 % of its people are coded at home
■ Register, Netherlands 2023	full matrix, 342 municipalities	jobs of employees	none	pairs to the nearest 100 jobs; place of work modelled. The 2014 table has only 40 × 40 regions of 1970
■ Operator trips, Spain 2023	full matrix, 3,792 districts	activity at both ends	hour of departure	trips of one operator's customers, expanded. "Work" includes study
■ Census, Serbia 2022	none: 4 nested bands	work; education as one number	none	"daily migrations". A migrant leaves the settlement, not the municipality
■ People-flow panel, Japan 2019 to 2021	none: 4 nested rings	none: presence	month, day type, day or night	"From-To". It holds no origins, and it is not operator data
■ OSM public GPS traces	trajectories of contributors	none	timestamps	200,000 points. 2,395 minutes of recording

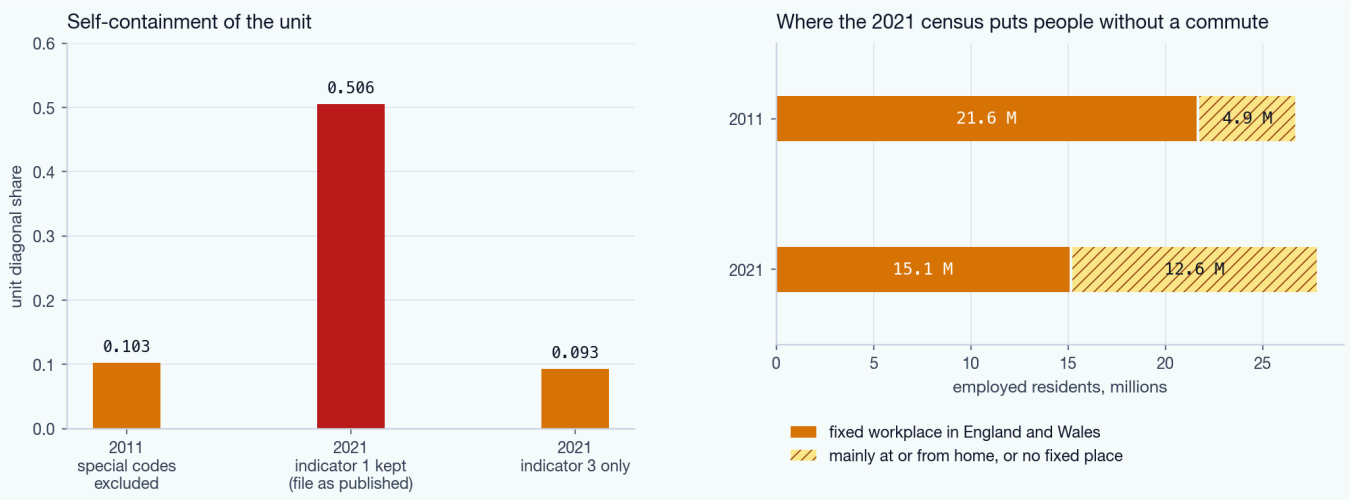
Eurostat statistics from mobile network operators: a published method and open reference code (Multi-MNO), no cross-country dataset.

The sources on six axes

Source	Spatial resolution	Temporal lag	Semantic depth	Licence	MAUP sensitivity	Cost to compute
■ Census pairs, England and Wales	7,264 polygons of 5,000 to 15,000 residents	2.6 years (census March 2021, file October 2023)	work; place-of-work indicator of 4 categories	OGL v3	measured: scale gives 19 % of the score at MSOA, 69 % at district level	1.8 M pairs; permutation 20 ms; contiguous partition 0.2 s
■ Census bands, Serbia	168 polygons, 4 destination bands	1.8 years	work; education	SORS terms	not testable, one level; 8 units are 0 by definition	seconds
■ Register pairs, Netherlands 2023	342 municipalities; pairs to 100 jobs	2 years; provisional	jobs of employees; place of work modelled	CC BY 4.0	measured: scale 64 %, contiguity 27 % of the COROP score	10,000 pairs; Coombes-Bond 119 s
■ Operator trips, Spain 2023	3,792 districts	a file per day since 2022; method change July 2025	activity at both ends; hour of departure	ministry open-data licence	measured: scale 31 %, contiguity 44 % of the Coombes-Bond score	190 MB a day; Coombes-Bond 46 min
■ Presence panel, Japan	1 km mesh, 6,763 cells	1 month while it ran; ends December 2021	no purpose; 3 × 3 time slots; 4 residence rings	Gov. of Japan Standard Terms 2.0	regular grid; class shares move with the threshold: mixed 17 to 65 %	1.07 M rows; 24 HDBSCAN runs 4 s
■ OSM GPS traces	points, to H3 resolution 8	none at upload; median year 2014 to 2022	position, time	ODbL	not measured; 50 % of points in one cell	40 requests per window
■ GHS-POP, GHS-SMOD grid	100 m, 1 km	epoch 2030 is a projection	population; 8 settlement classes	CC BY 4.0	measured: $r = -0.28$ by area, -0.02 by population	75 MB tile; 3 s

Test 1. Read the code list

In 2021 people who work at home or have no fixed workplace are rows with workplace = residence. They sit on the diagonal of the matrix.



Matrix	people	diagonal
2011, special codes out	21,625,060	0.103
2021, as published	15.1 M + 12.6 M	0.506
2021, indicator 3 only	15,085,108	0.093

The file as published says that local working rose fivefold. With the same treatment in both years it fell from 0.103 to 0.093.

Census day was 21 March 2021, in a lockdown. The 2021 matrix is used here to study coding and method, not commuting.

Test 2. Shuffle. The null has a closed form

Keep the sizes of the areas, permute the labels of the units. What survives is produced by the number and sizes of the areas alone.

$$E[SC_0] = d + (1 - d) \cdot h$$

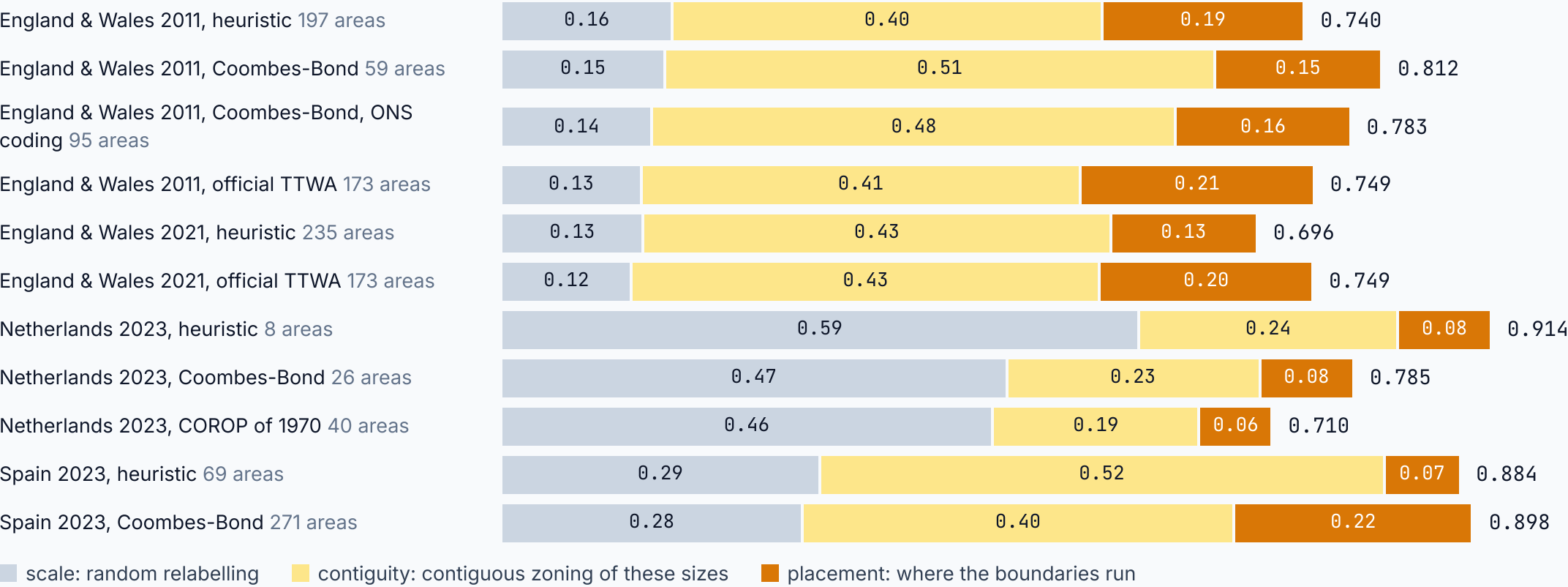
d: share of flows that stay in their unit. $h = \sum n_a(n_a - 1) / (n(n - 1))$: chance that two units share an area. $1/h$ is the effective number of areas.

England and Wales 2021	MSOA	district
units	7,264	331
d	0.093	0.551
areas, same rules	235	9
self-containment	0.695	0.947
null, 1,000 permutations	0.130 ± 0.001	0.657 ± 0.008
null, closed form	0.130	n/a
share of the score that is scale	19 %	69 %



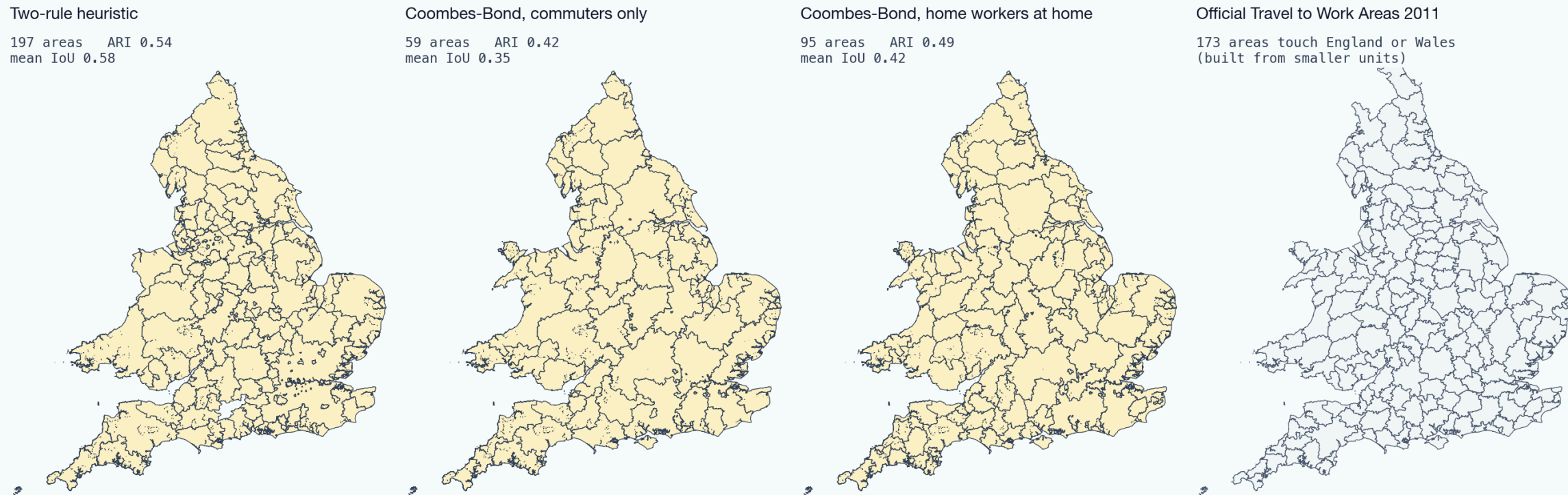
Most of the score is scale and contiguity

A second null keeps contiguity and the sizes: a recombination chain on spanning trees, 100 contiguous partitions per row, sizes within 5 %. What is left is what the position of the boundaries adds.



Test 3. Compare the map, not the count

One published criterion, four maps. Same algorithm, same parameters: 59 areas from commuters only, 95 when people who work at home or have no fixed workplace are counted at home, as ONS counted them. The official map has 173.



The error matrix, counted in people

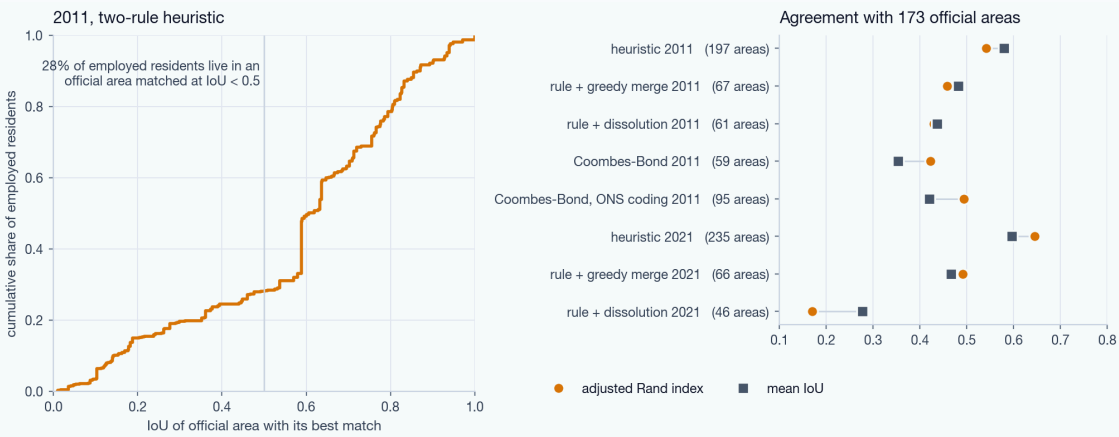
$C[a, b]$ = employed residents who are in delimited area a and in official area b. Everything else is read from it.

$$IoU(a, b) = C[a, b] / (C[a, \cdot] + C[\cdot, b] - C[a, b])$$
$$omission(b) = 1 - C[a^*, b] / C[\cdot, b]$$
$$commission(b) = 1 - C[a^*, b] / C[a^*, \cdot]$$

a^* is the delimited area with the highest IoU for official area b. Adjusted Rand index and V-measure come from the same table.

Construction	areas	ARI	mean IoU	by km ²
heuristic	197	0.54	0.58	0.44
rule by greedy merge	67	0.46	0.48	0.37
rule by dissolution	61	0.43	0.44	0.33
Coombes-Bond, commuters only	59	0.42	0.35	0.34
Coombes-Bond, home workers at home	95	0.49	0.42	0.46

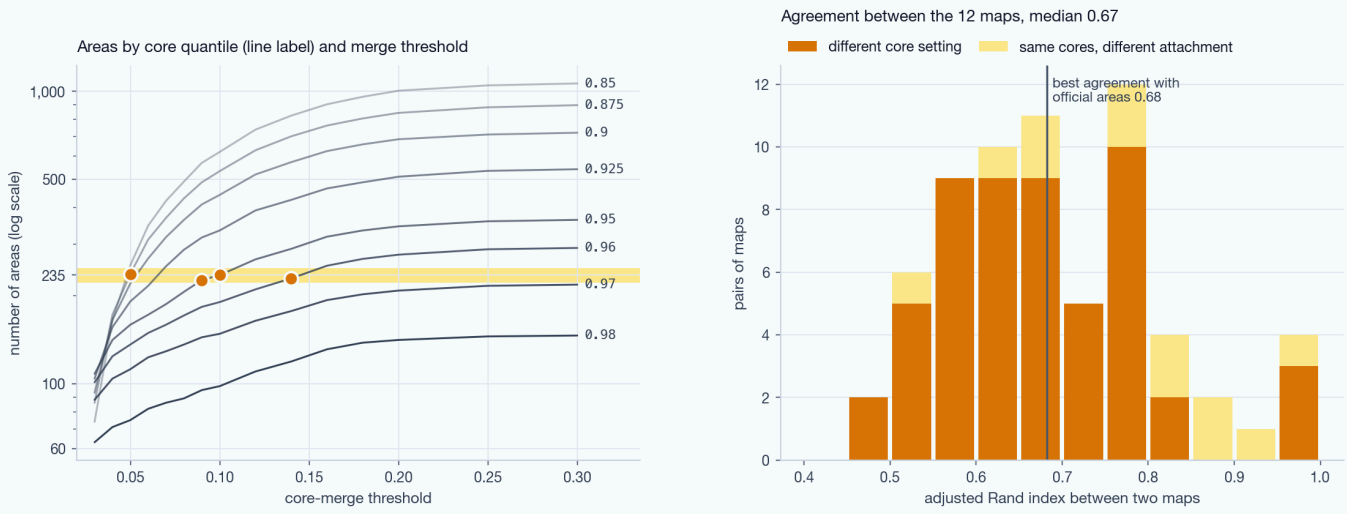
2011 matrix. On the 2021 matrix dissolution ends with 46 areas, one of them with 45 % of employed residents.



Naive repairs of the rule move the map away from the official one: under greedy merge the London area holds 28 % of England and Wales. Coding home workers as ONS did moves Coombes-Bond from ARI 0.42 to 0.49; the heuristic, nearest in count, still agrees best.

The same count from different rules

The number of areas is fixed by two of the three thresholds. 4 settings of those two give about 235 areas.

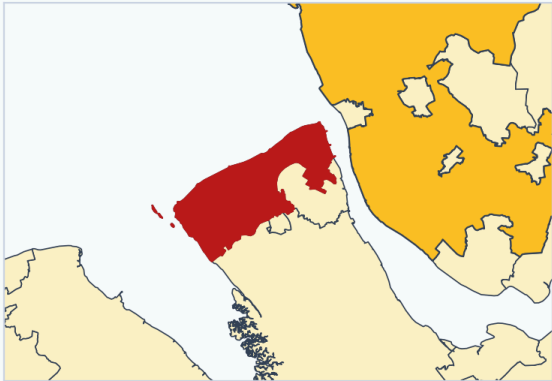


12 maps with 223 to 247 areas	min	median	max
agreement with each other, ARI	0.46	0.67	0.98
agreement with official areas, ARI	0.54		0.68
self-containment	0.641		0.737
areas over the whole grid of 120	63		1,063

Where the boundaries diverge: six cases, one by one

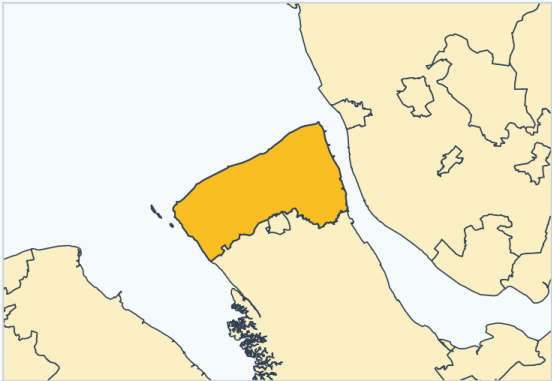
Chosen by rule from the 2021 matrix, not by eye. The next six slides take them in turn.

1 Attachment without adjacency



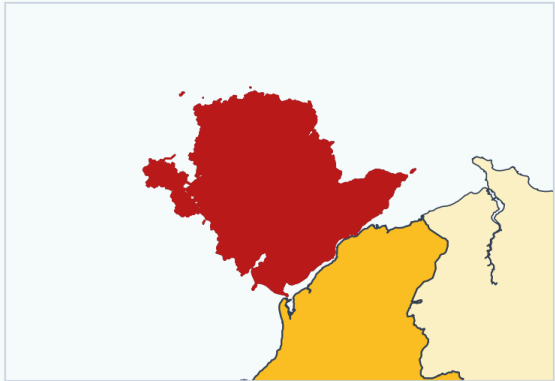
18 units with 34,324 employed residents join the Liverpool area by flow. The nearest unit of that area is 1.1 km away, across water.

2 The same window after contiguity repair



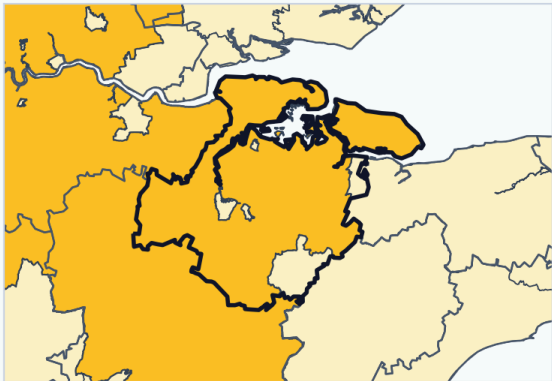
The fragment goes to the adjacent area with the strongest flow across the shared border (here the area around Wirral).

3 An island the repair cannot place



9 units of Isle of Anglesey touch no unit of another area. Polygons end at the shore; bridges are not adjacency.

4 Official area inside a larger one: Medway



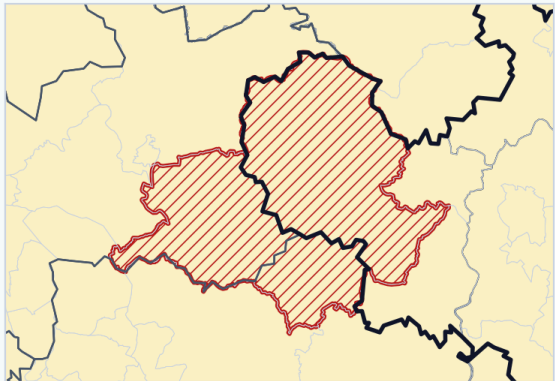
Black: official boundary. The matching delimited area (darker) holds 93% of its employed residents and 94% of the match lies outside. IoU 0.06.

5 Official area cut into 5: Slough and Heathrow



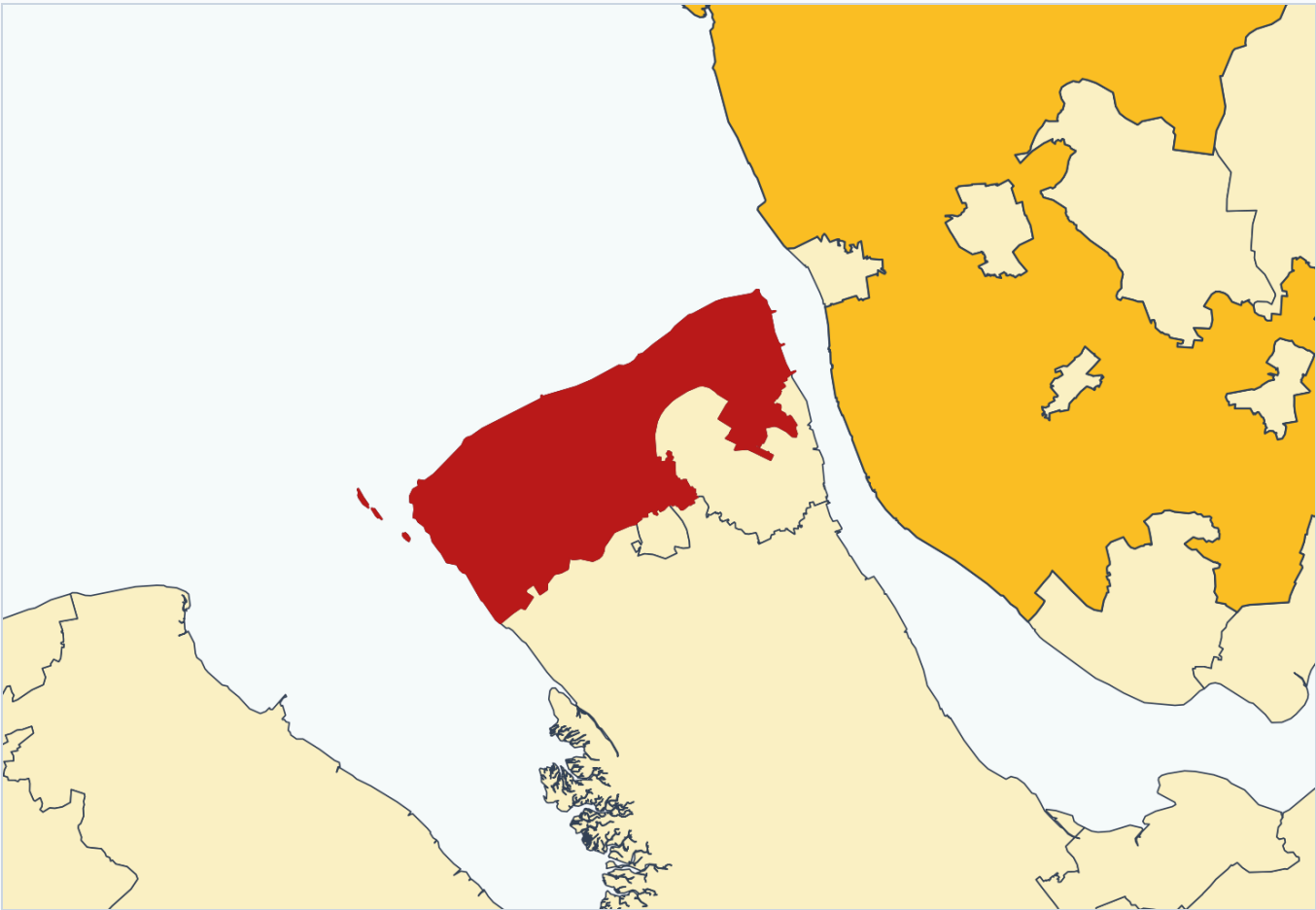
The best-matching delimited area (darker) holds 44% of its employed residents. IoU 0.44.

6 A unit the official boundary cuts: Monmouthshire 005



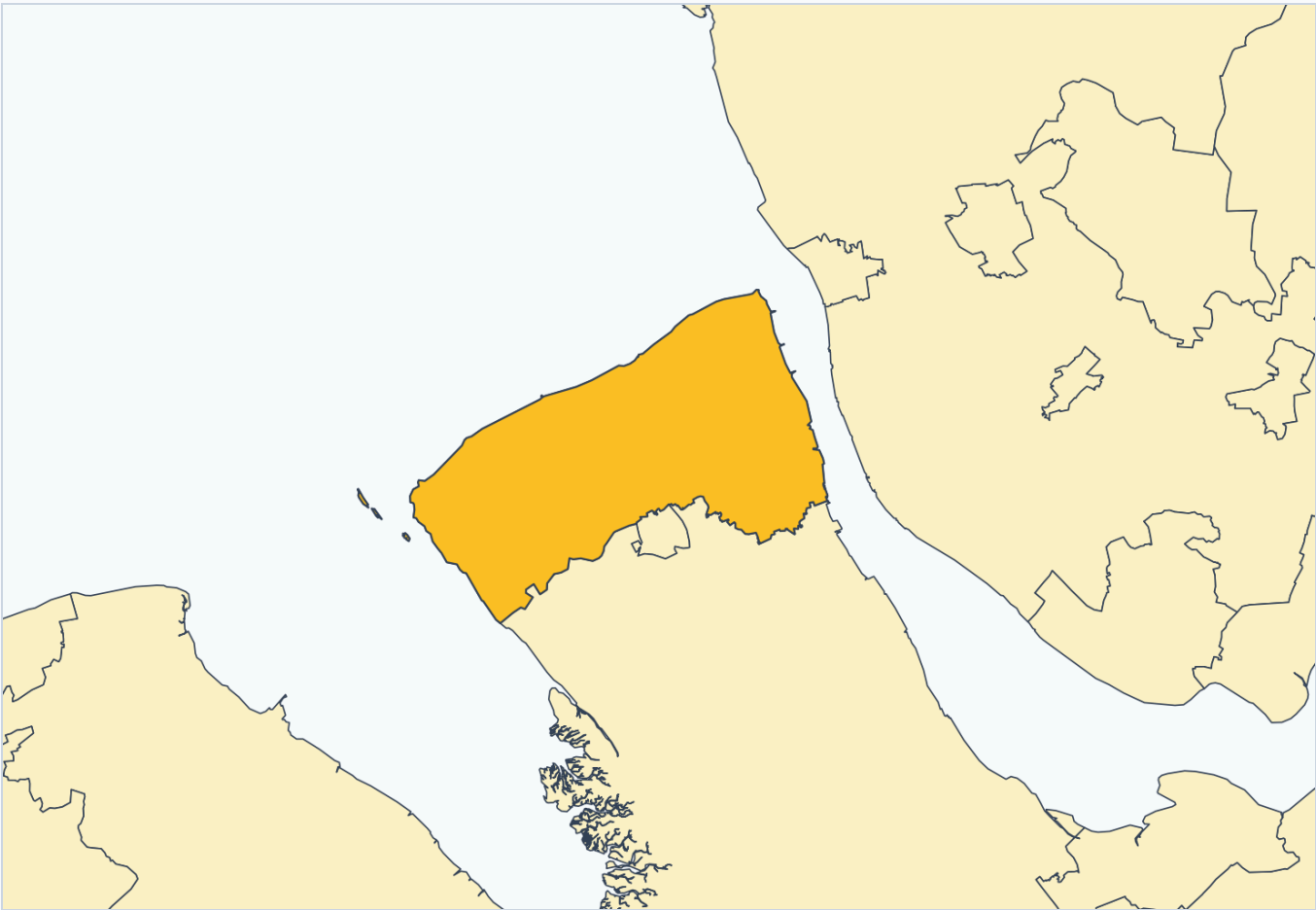
Official areas are built from smaller blocks. 41% of this unit's blocks lie in its plurality area; 332 of 7,264 units are cut this way.

Case 1. Attachment without adjacency



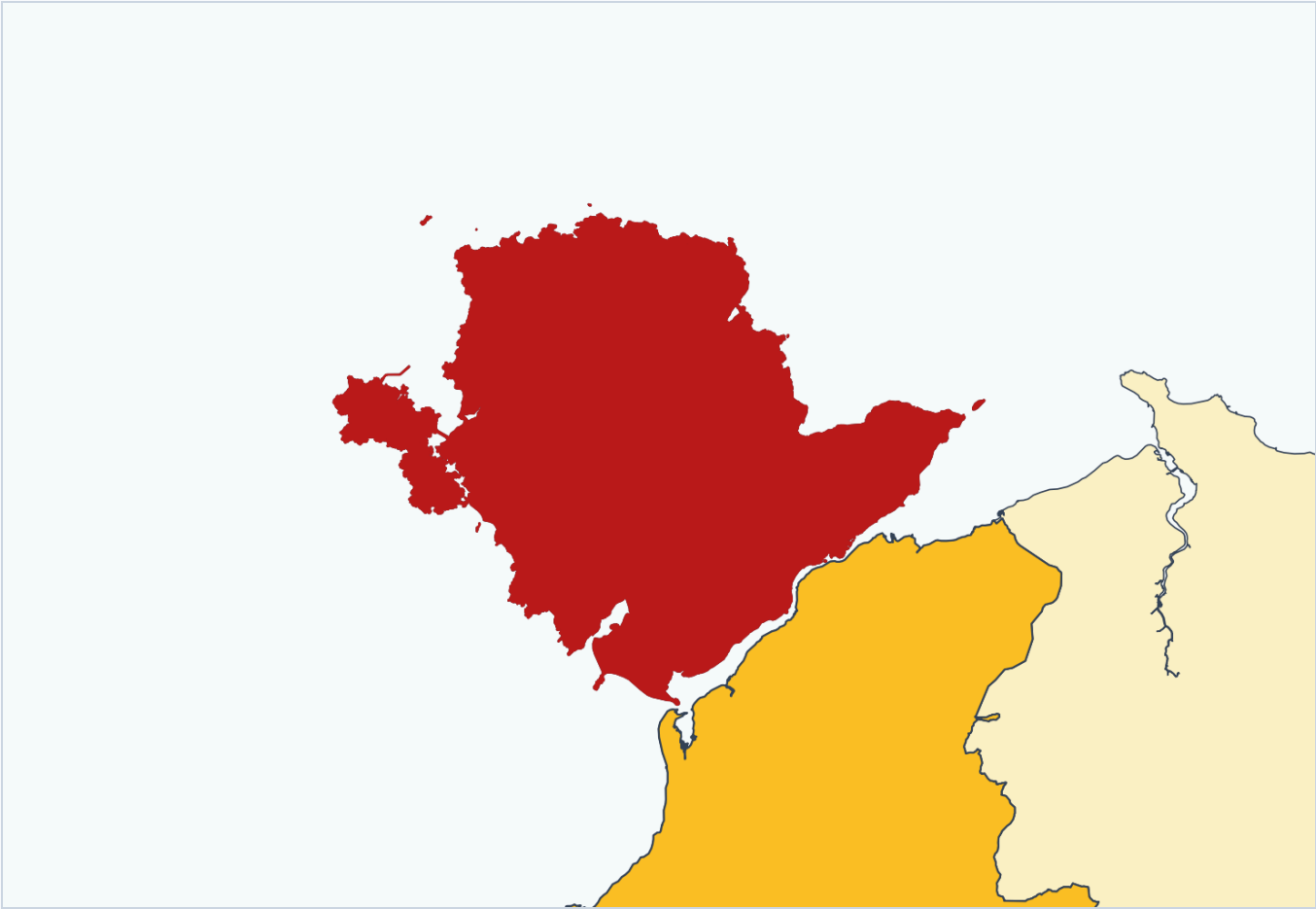
Rule	A unit joins the area that receives the largest share of its outgoing commuters, if that share is 15 % or more. The rule looks at flows only.
Result	18 units with 34,324 employed residents join the Liverpool area. The nearest unit of that area is 1.1 km away, across water.
Extent	26 of 235 areas are in more than one piece; 79 units are cut off.
Reading	The polygons do not cross the river. A tunnel does. By flows the fragment belongs where the rule put it.

Case 2. The repair



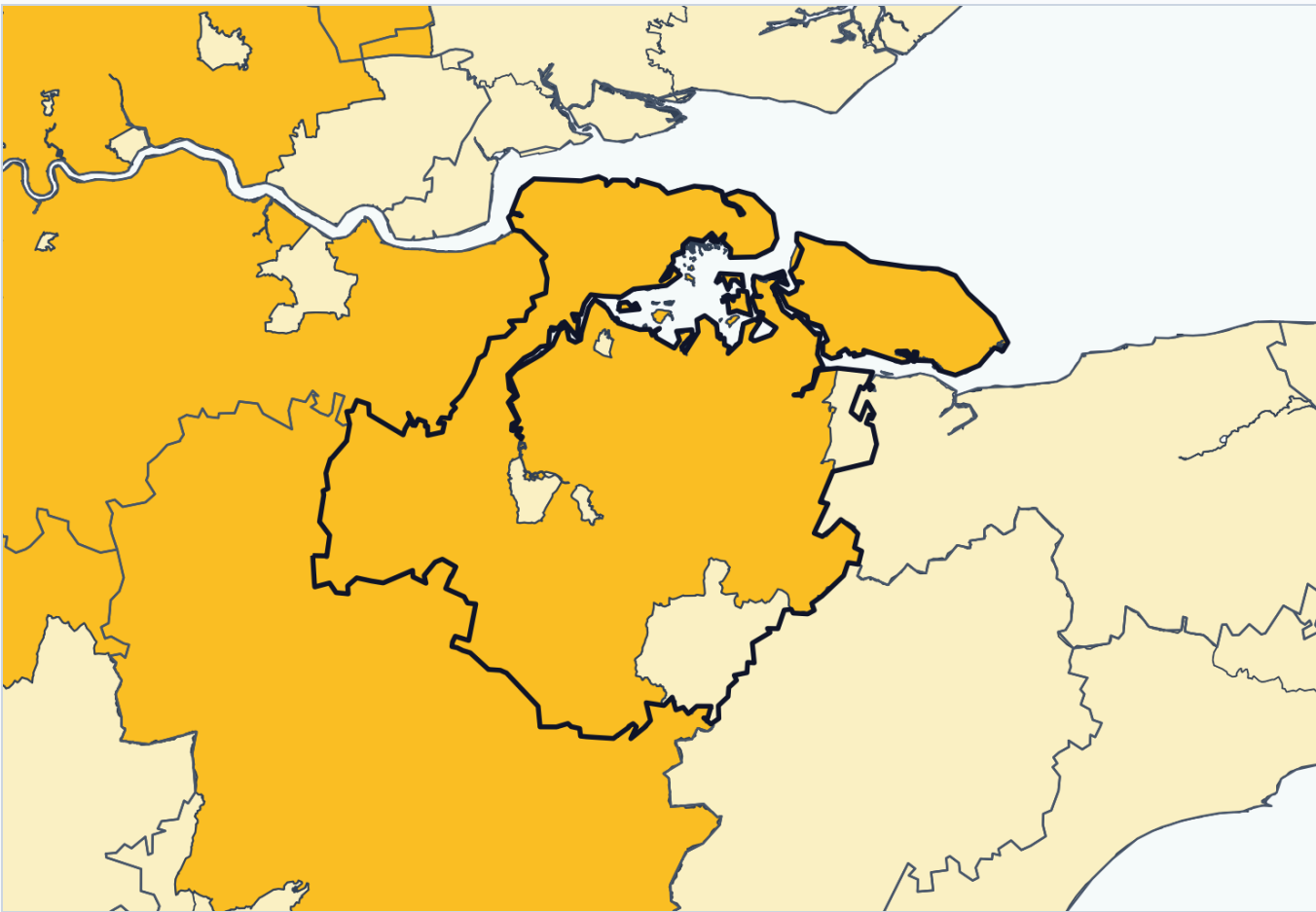
Rule	In each area the piece that holds the core stays. Every other piece goes to the adjacent area with the largest two-way flow across the shared border. Adjacency is queen adjacency of the polygons.
Result	The fragment now belongs to the area around Wirral.
Cost	Self-containment moves from 0.695 to 0.696.
Reading	The repair gives a map that can be drawn. In this case it replaces a true functional link with a polygon rule. Both versions are in the tables.

Case 3. An island the repair cannot place



Rule	A piece with no neighbour in another area has nowhere to go. It stays and is reported.
Result	9 units of Isle of Anglesey, 17,948 employed residents.
Extent	3 areas keep such a piece after the repair, 13 units in all.
Reading	Polygons end at the shore; two bridges do not count as adjacency. The list of islands is an output of the pipeline, not a defect to hide.

Case 4. An official area inside a larger one



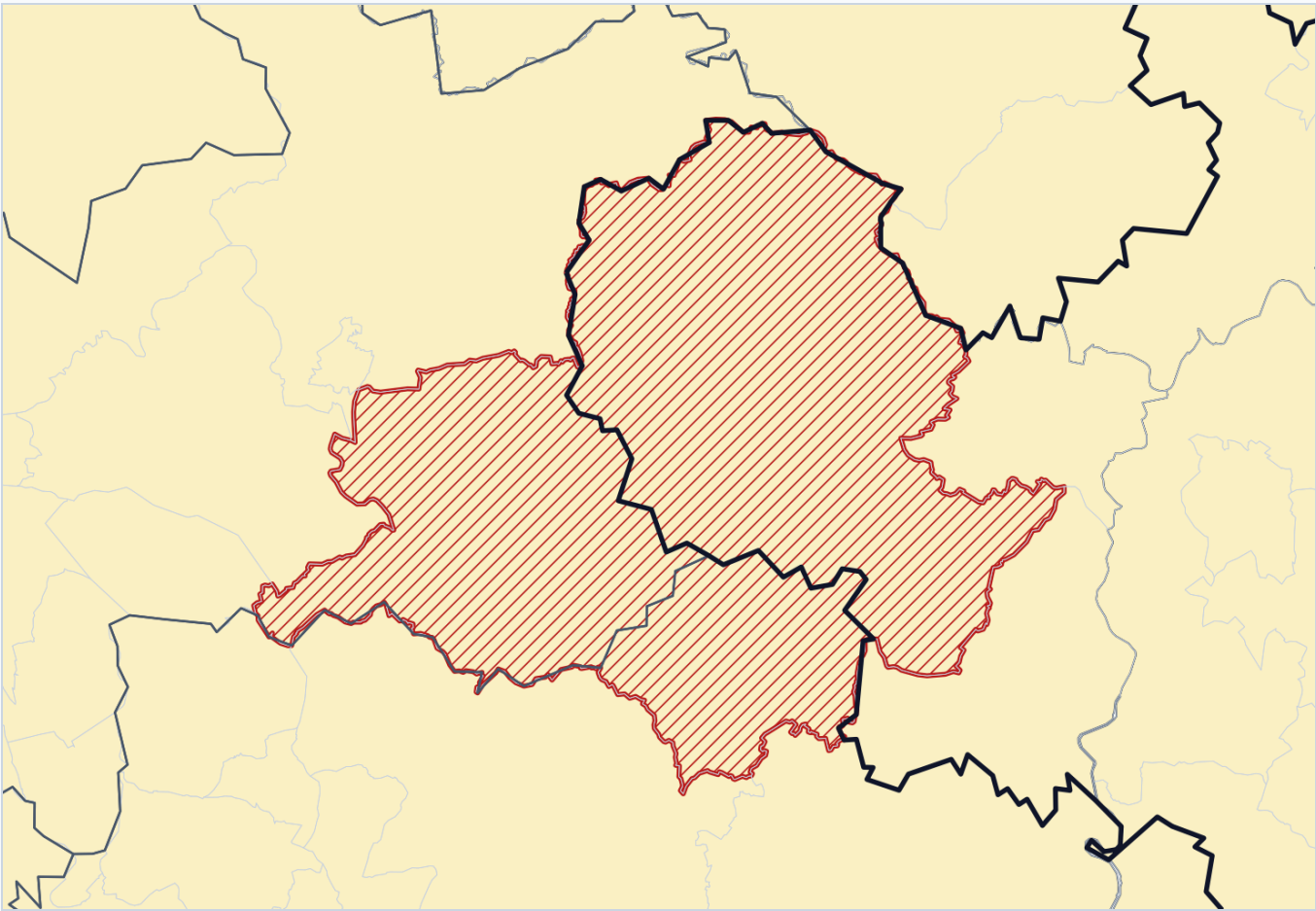
Official	Medway, 166,407 employed residents (black line).
Match	The delimited area that starts at City of London (darker amber).
Error matrix	IoU 0.06. Omission 7 %: nearly all of the official area is inside the match. Commission 94 %: nearly all of the match is outside it.
Reading	The heuristic has no lower bound on self-containment of what it attaches. One strong core takes its whole commuter belt, other labour markets included.

Case 5. An official area cut into several



Official	Slough and Heathrow, 401,486 employed residents (black line).
Match	5 delimited areas hold parts of it. The best one holds 44 % of its employed residents: omission 0.56.
Error matrix	IoU 0.44. Commission is 0: the match lies inside the official area.
Reading	Several cores pass the inflow quantile here and do not exchange 10 % of their outflow, so they are not merged. The count of areas rises where employment is polycentric.

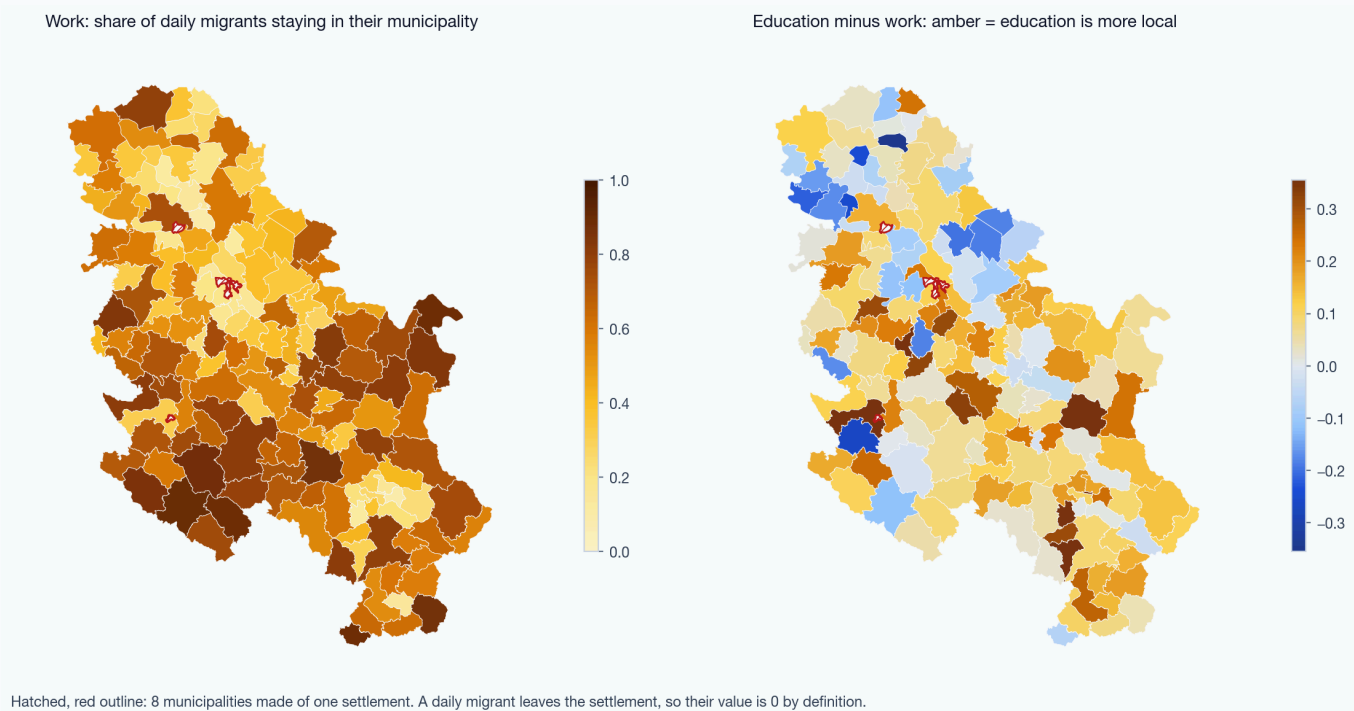
Case 6. A unit the official boundary cuts



Cause	Official areas are built from blocks smaller than the units of the open matrix.
Result	Monmouthshire 005: 41 % of its blocks lie in the official area it is assigned to (hatched unit, black line).
Extent	332 of 7,264 units are cut by an official boundary.
Reading	No partition of these units can reproduce the official map. Agreement values carry this error and cannot reach 1. This is the zoning problem at the level of the unit.

Serbia. Bands instead of pairs

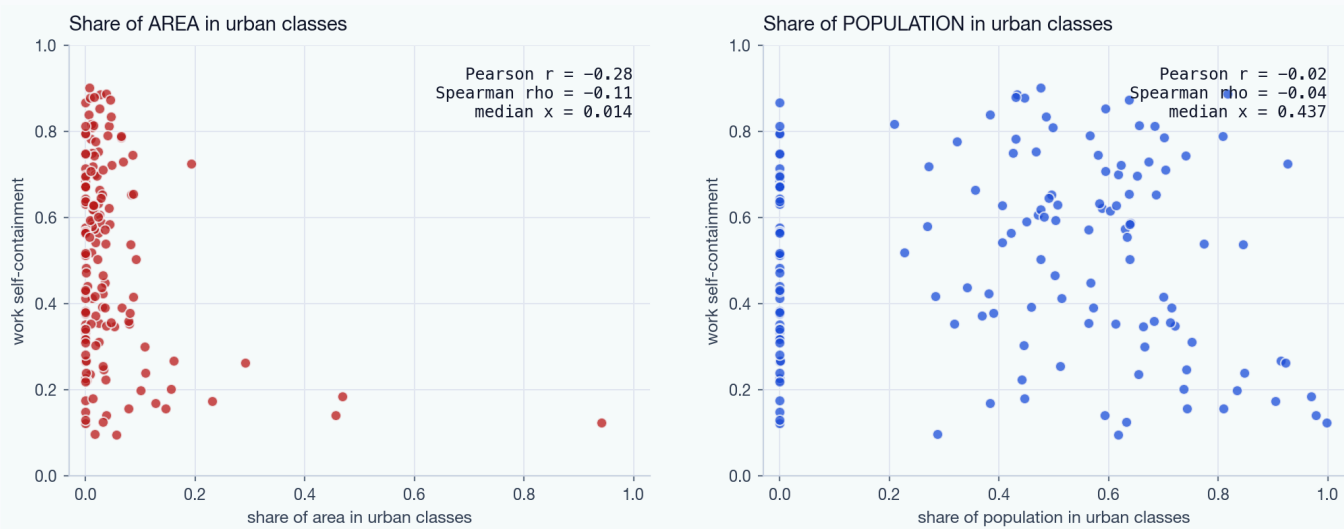
795,779 daily migrant workers in 168 municipalities. Delimitation is not possible. Band self-containment is, if the definition is respected.



	value
stay in their municipality	50.3 %
median, work	0.53
median, education	0.65
median of municipal differences	+0.08
correlation, work and education	0.80
municipalities of one settlement: value 0 by definition	8
polygons joined, one to one	168

Weight the grid by people, not by pixels

Degree of urbanisation of a municipality from GHS-SMOD classes, weighted by GHS-POP population. The area-weighted share is the control.

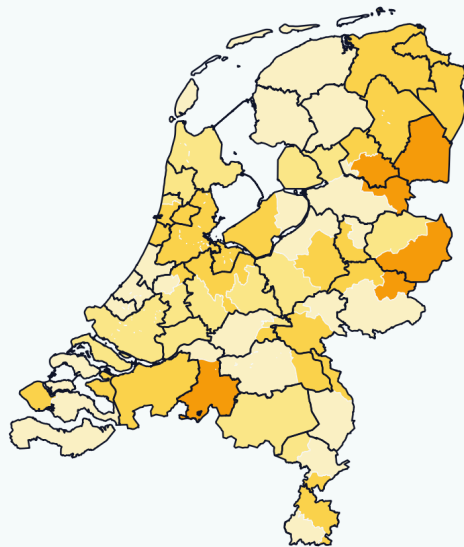


Work self-containment against urban share	Pearson	Spearman
share of area	-0.28	-0.11
share of population	-0.02	-0.04
Education minus work		
share of area	+0.17	
share of population	+0.03	
median share of area in urban classes	0.014	

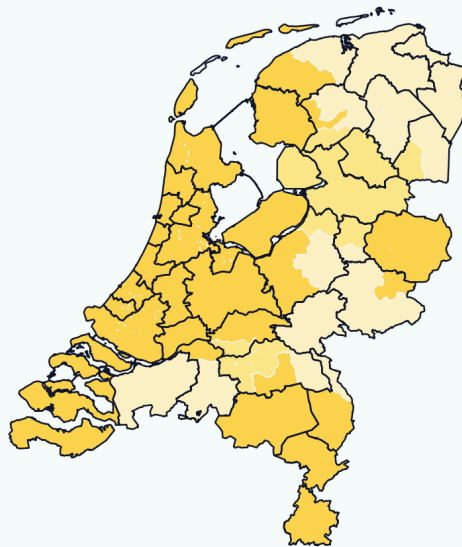
Netherlands. Do the regions of 1970 still hold their commuters?

The 2014 table published pairs only between the 40 COROP regions, so it could not test them. Its successor publishes pairs between 342 municipalities.

Coombes-Bond: 26 areas, ARI with COROP 0.50



Two-rule heuristic: 8 areas, ARI 0.14



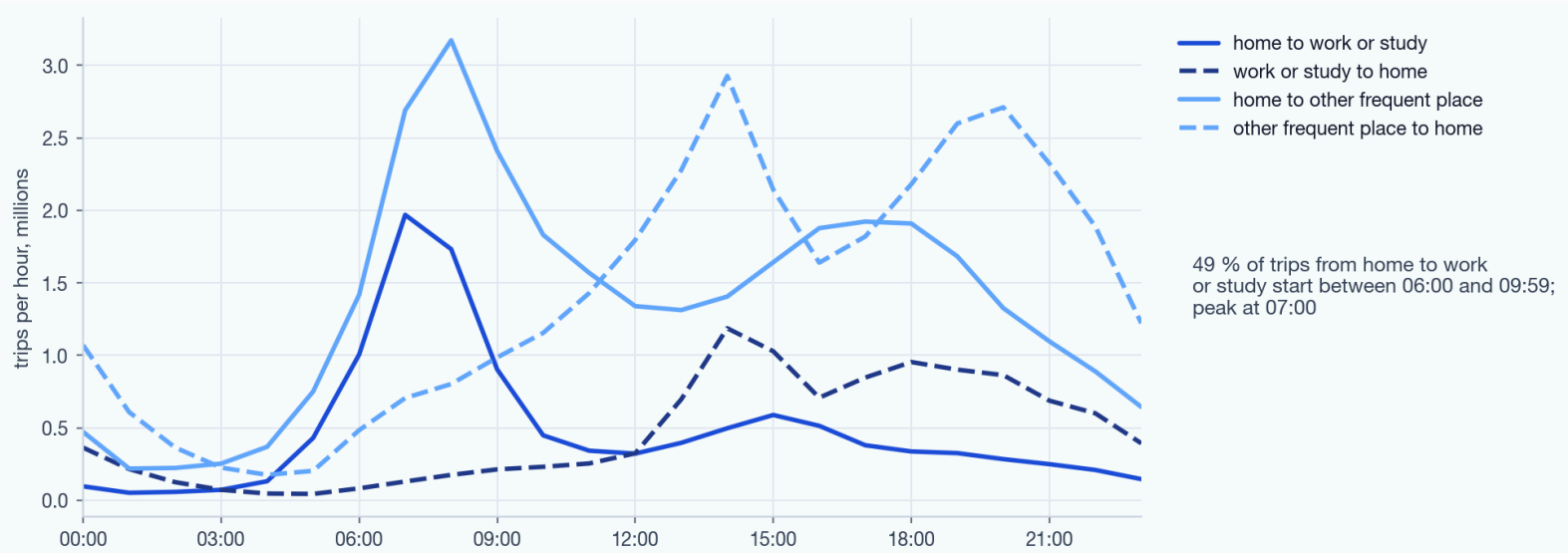
COROP regions: 20 % hold 75 % of residents' jobs



December 2023	value
COROP regions, median self-containment	0.66
COROP regions valid under the TTWA rule	45 %
Coombes-Bond on municipal pairs: areas	26
its agreement with COROP, ARI	0.50
placement term, COROP	0.065
placement term, Coombes-Bond	0.085
jobs retained by the rounded pairs	95.6 %

Spain. Pairs, purpose and the hour in one open file

Trips between 3,792 districts by hour and by the activity at both ends, from the data of one operator, expanded to the population.



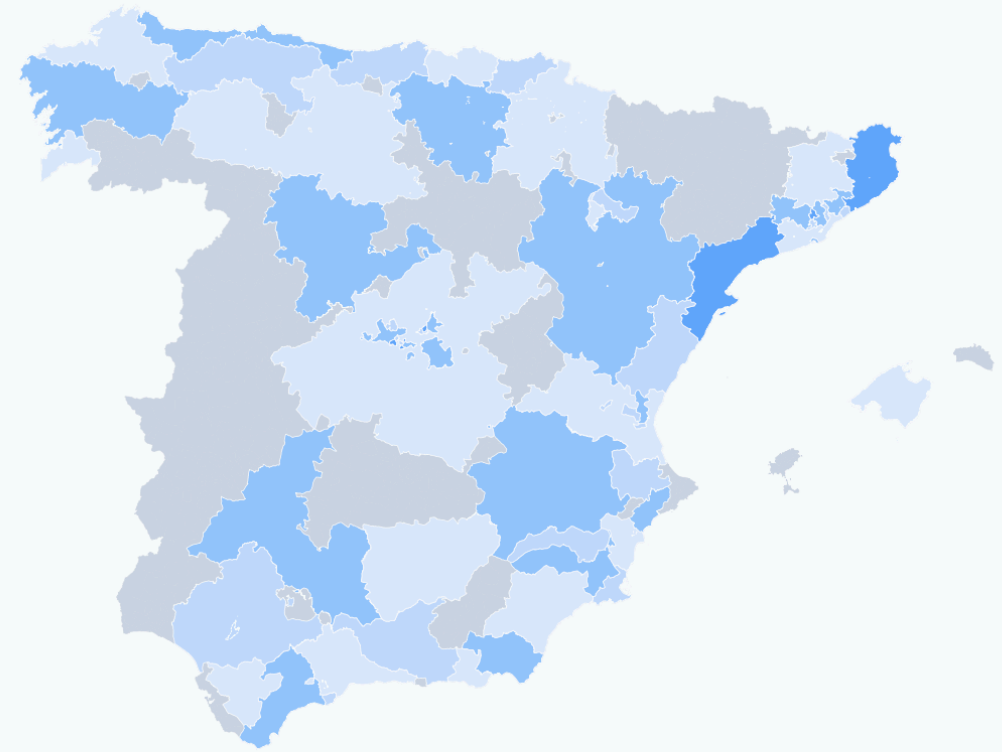
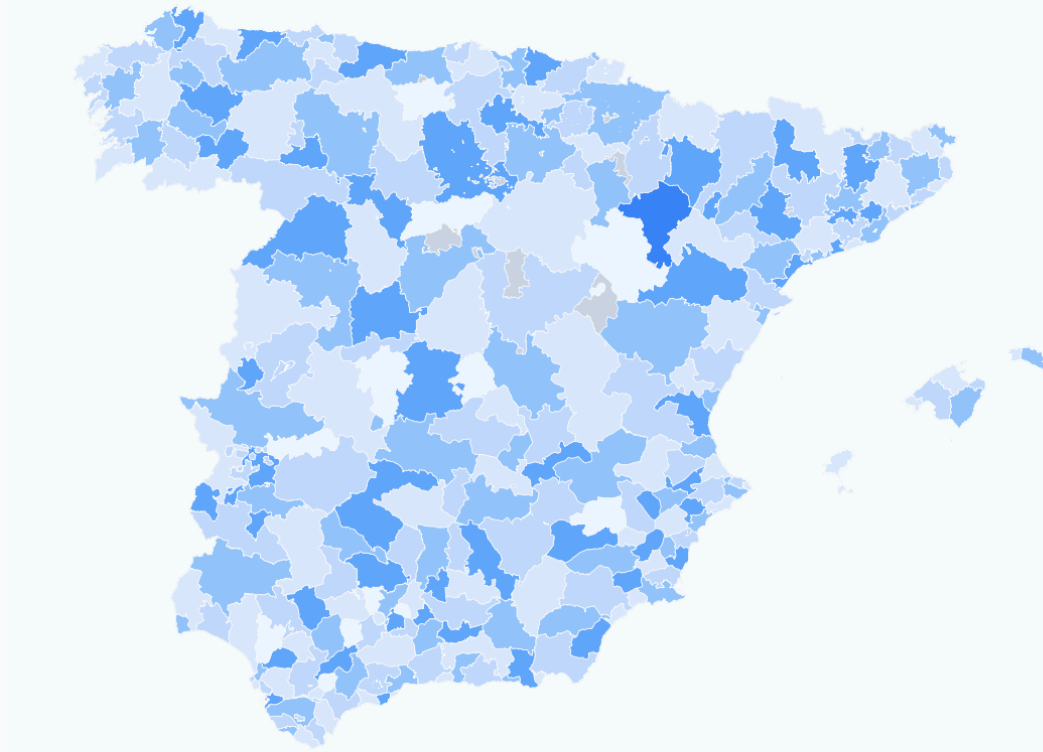
Weekdays, October 2023		value
trips a day, all purposes		143 M
home to work or study, destination not flagged as study		11.31 M
of these, starting 06:00 to 09:59		49 %
Coombes-Bond areas		271
heuristic areas		69
zones the heuristic leaves out		751 (14 % of trips)
agreement of the two maps, ARI		0.59

Spain. The same two constructions, a different country

The heuristic, with the parameters set on England and Wales, leaves 14 % of trips without an area. Coombes-Bond assigns every district that sends trips and makes 271 smaller areas.

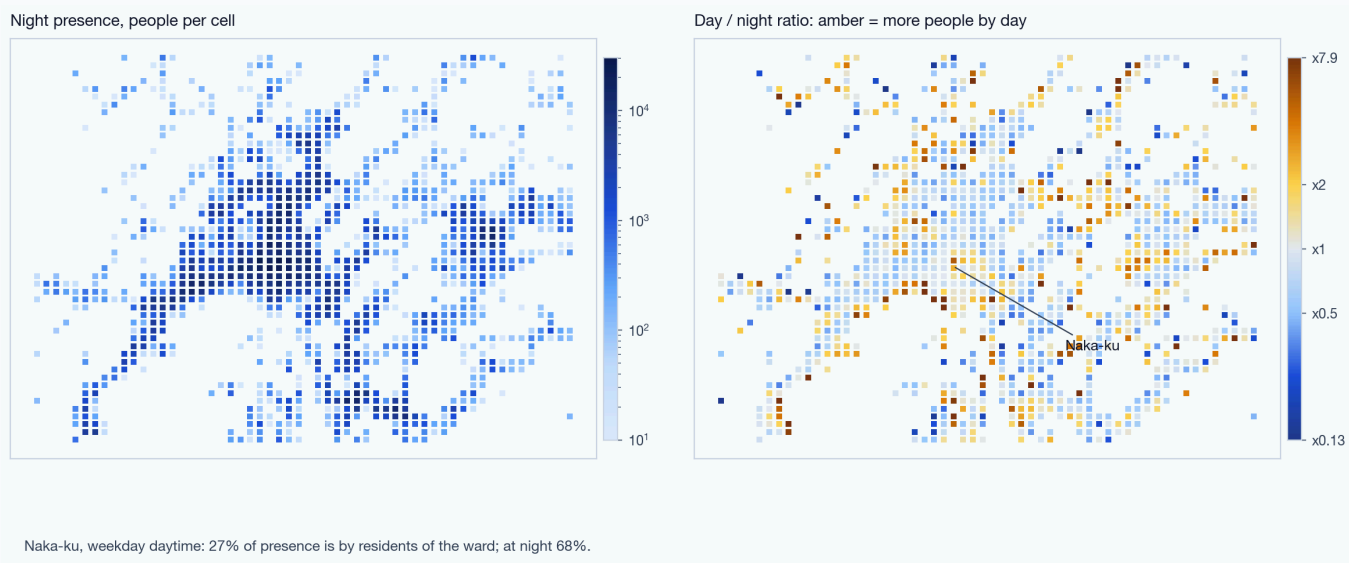
Coombes-Bond: 271 areas

Two-rule heuristic: 69 areas



Japan. Time of day, and nothing else

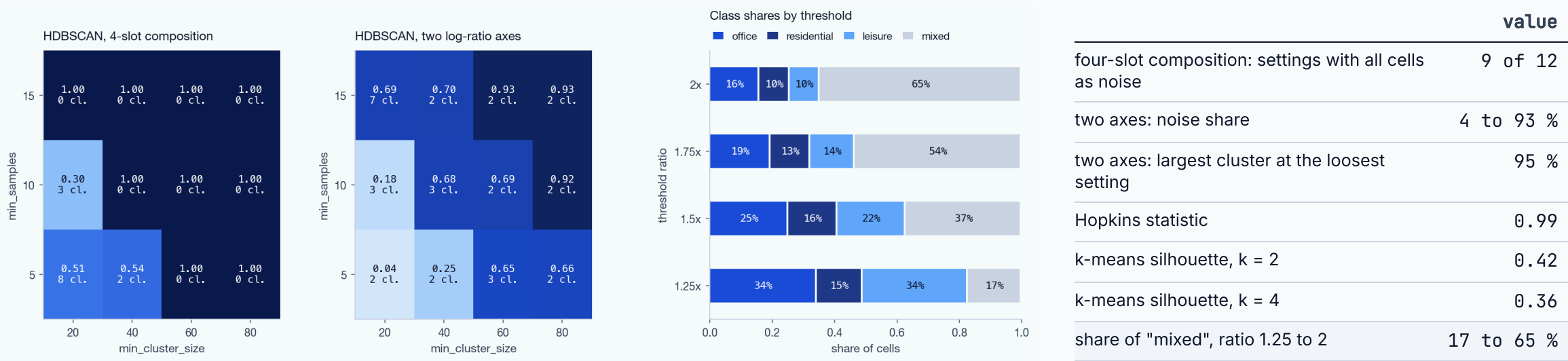
Presence per 1 km cell by month, day type, day or night. A panel of smartphone applications, expanded to population by the provider. Not operator data.



	value
Naka-ku, weekday day: presence by residents of the ward	27 %
the same at night	68 %
cells with more people by day	43 %
yearly totals 2019 to 2021 agree within	0.11 %
local share, weekday day, 2019	0.632
2020	0.677
2021	0.681

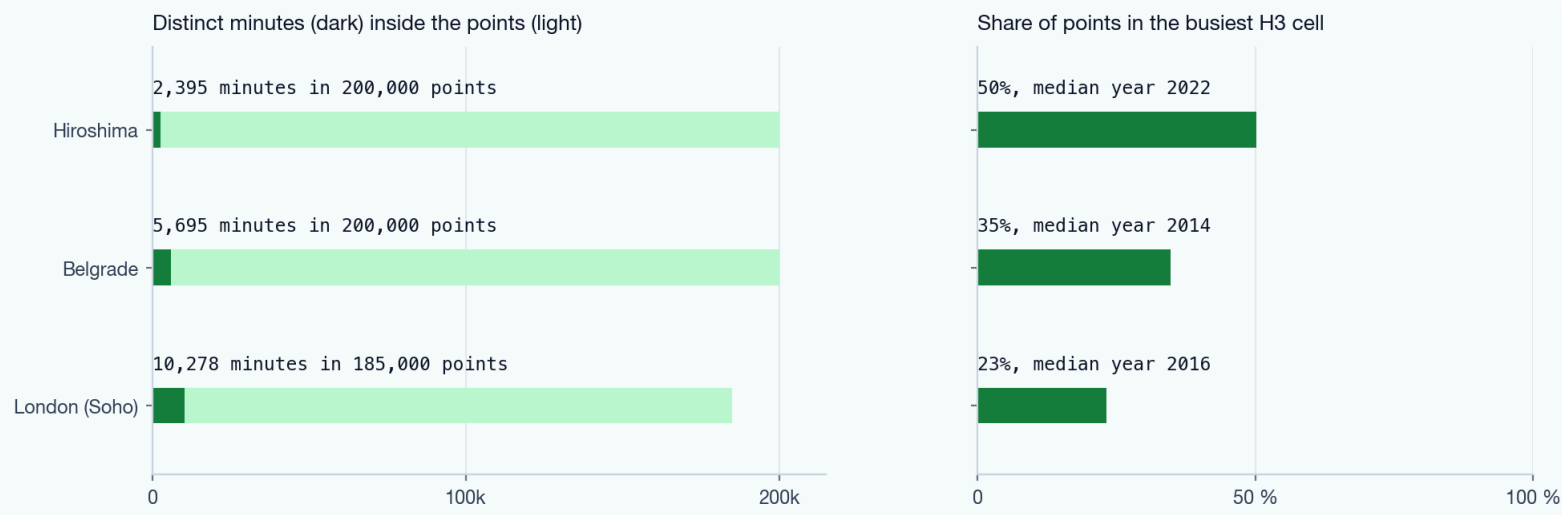
Test 4. Ask whether there are classes before naming them

Studies of phone data cluster temporal profiles and name the clusters as land uses. 4,011 Hiroshima cells, four time slots: HDBSCAN finds no classes, and rule-based classes return the share the threshold asks for.



OpenStreetMap traces. Count minutes, not points

200,000 points is the limit of the request: 40 pages of 5,000.



Window	minutes	median year	busiest cell
Hiroshima	2,395	2022	50 %
Belgrade	5,695	2014	35 %
London, Soho	10,278	2016	23 %

Version 1 compared traces with daytime presence and with mapped places. 11 shared cells give no power, and the test is withdrawn. What remains is a statement about the source: a few thousand minutes of recording per window.

Deck v1, audited

56 numbers in the first version of these slides. 46 reproduced within the rounding of the slide, 10 corrected. Three statements of fact corrected.

What the deck said	What the pipeline gives	Kind of error
District level: 0.961 observed, 0.798 random, "80 % is arithmetic"	0.947 and 0.657; 69 %	computed before people coded at home were removed
Most open Dutch region: Delfzijl 0.45	Zaanstreek 0.45; Delfzijl 0.54	right number, wrong name
Most closed: Zeeuwsch-Vlaanderen 0.88	Zuid-Limburg 0.88; Zeeuwsch-Vlaanderen 0.85	right number, wrong name
Education-work difference against urban area share: $r = -0.28$	-0.28 belongs to work self-containment; for the difference $+0.17$	right number, wrong quantity
HDBSCAN: 100 % noise	at 9 of 12 settings on composition; never on the two axes	condition left out
Work against education: $r = 0.82$	0.80; 0.85 with the 8 structural zeros	no variant gives 0.82
Belgrade, traces against places: $+0.67$	$+0.72$	cell table regenerated since
235 areas, then 67 after the rule	235 is 2021; 67 comes from the 197 of 2011. 2021 gives 66	two years in one sentence
Japanese data are operator data of NTT docomo	a panel of smartphone applications; provider Agoop	fact
Only Japan publishes time of day	Spain publishes hourly operator matrices	fact
Dutch table discontinued after 2014	successors exist; the current one has municipal pairs	fact

Which source for which question

Question	Source that can answer it	Check before use
Delimitation, commuting structure	■ census or register pairs at fine units; ■ operator trip matrices	codes for people without a commute; unit of count; both baselines; contiguity; census date
Self-containment where no pairs exist	■ destination bands	definition of the migrant; units of one settlement
Trips and presence by time of day	■ operator trip matrices; panel products	who is in the sample; expansion; normalisation; suppression threshold; what "from" means
Urbanisation of a unit	■ population grid with settlement classes	weight by population; epoch of the raster
Routes of single recordings	■ OSM public traces	distinct minutes; concentration; age of traces

Four tests that need only the open file

Read	the code list. Where do people without a trip land in the matrix?
Shuffle	$d + (1 - d) h$, then contiguous partitions of the same sizes from a recombination chain.

Compare	constructions on the placement term, and maps through the error matrix, in people.
Test	for clusters before naming classes; state the thresholds.

Limits

- Three kinds of count: persons at a census, employee jobs with a modelled place of work, expanded trips. Compare partitions within a matrix, not magnitudes across countries.
- The 2021 census day was a lockdown day. Comparisons with the official map use 2011.
- Official areas are built from smaller blocks; 332 units are cut.
- The recombination chain targets the spanning-tree distribution, which favours compact areas. It is not a uniform sample.
- The official production code was not run, and the official map was cleaned by a secondary process. LabourMarketAreas is the open implementation of the published method.
- The linear form of the validity trade-off is an assumption.
- Dutch pairs are rounded to 100 jobs and keep 95.6 % of jobs; all figures provisional.
- Spanish trips mix work with study, come from one operator and cover five weekdays. The size thresholds were set for persons.
- GHSL epoch 2030 is a projection: 6.87 million people against about 6.65 million in the census.
- Who is in the Japanese panel is not published. Cells under 10 people are absent.

Next

Coombes-Bond on the 2021 matrix and on the Dutch series 2021 to 2024. The package's contiguity fine-tuning in the decomposition. The Spanish matrix of recurrent mandatory mobility, which counts persons. Japanese census pairs for Hiroshima.

Run it

```
git clone https://github.com/m-erts/urban-functional-maps
conda env create -f environment.yml
make env-r # R, for Coombes-Bond
python scripts/fetch_data.py
make all
```

What keeps the numbers honest

One parameter file. One file of results. Documents rendered from templates: no number is typed.

A frozen snapshot is the regression target of the tests.

Reconciliation tests: sums equal published totals, joins are one to one, structural zeros are explained.

Input files listed with address, licence and SHA-256.

The run repeated in a fresh environment from the lock file gave the same values.

Repository

github.com/m-erts/urban-functional-maps

Paper

[docs/paper/paper.pdf](#)

Licences

Code MIT. Text, figures, slides CC BY 4.0.

Data

Office for National Statistics, OGL v3.0. Statistics Netherlands (CBS), CC BY 4.0. Basado en datos abiertos Ministerio de Transportes y Movilidad Sostenible. Statistical Office of the Republic of Serbia. GeoSrbija. MLIT Japan, provider Agoop. European Commission JRC, GHSL, CC BY 4.0. © OpenStreetMap contributors, ODbL.

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